13 Hydrogen

The dihydrogen (H_2) available in the most abundance in the world comes third in the proportion on the surface of the earth. It is seen as the largest source of energy in future.

Symbol : H

Atomic number : 1

Atomic weight : 1.008 u

- **Position of Hydrogen in periodic table**

Similarity with Alkali metals			
No.	**Property**	**Hydrogen**	**Alkali metals**
(1)	Electronic configuration	$1s^1$	ns^1
(2)	Oxidation state	+1	+1
(3)	Reductant	strong	strong
(4)	Halogen elements and reaction with oxygen	stable halide and oxide	stable halide and oxide
Non-similarity with Alkali metals			
No	**Property**	**Hydrogen**	**Alkali metals**
(1)	Ionization Enthalpy	very high 1312 KJ/mole	very less Li = 520 KJ/mole
(2)	Physical state	Diatomic Gas	Solid
Similarity with Halogen			
No	**Property**	**Hydrogen**	**Halogen**
(1)	Oxidation state	−1 (Hydride)	−1
(2)	Atomic	Diatomic	Diatomic
(3)	Ionization enthalpy	1312 KJ/mole	F=1680 KJ/mole
(4)	Reaction with metals	Hydride e.g. NaH	Halide e.g NaCl

- **Non-similarity with Halogen**

No.	**Property**	**Hydrogen**	**Halogen**
(1)	Electronic configuration	$1s^1$	$ns^2\ np^5$
(2)	Colour	Colourless	Coloured
(3)	Oxidant	Weak	Strong

Thus position of hydrogen is not definite, so it is placed in the middle of first period.

● **Isotopes of Hydrogen**

No	Name and symbol	Atomic Number	Atomic mass	Number of proton	Number of neutron	Proportion	Stability and $t_{\frac{1}{2}}$
(1)	Protium $^{1}_{1}H$	1	1	1	0	Highest 99.98	Stable
(2)	Deuterium $^{2}_{1}H$	1	2	1	1	0.015 %	Stable
(3)	Tritium T, · $^{3}_{1}H$	1	3	1	2	T : $^{1}_{1}H$ $1 : 10^{17}$	Radio active $^{3}_{1}H \rightarrow {}^{3}_{2}He + {}_{-1}e^{0}$ $\frac{t_1}{2}$ = 12.33 year

The physical properties of these isotopes differ from each other due to large difference in atomic weight.

Their electronic configuration are same, so their chemical properties are similar, but have different rate of reaction.

1. Which element is seen as the largest source of energy in future ?

(A) Monoatomic gas (B) Diatomic non metal gas

(C) least weight of diatomic gas (D) triatomic gas

2. Which of the following possess electronic configuration similar to that of hydrogen ?

(A) strong oxidant elements (B) Noble gases

(C) Anhydrous substance (D) Strong reductant elements

3. Which property of hydrogen differ from strong reducing agent elements ?

P. Preparation of stable peroxide Q. $\Delta_i H$ R. physical state

(A) Q, R (B) only P (C) only Q (D) P, R

4. To which of the following reagent hydrogen acts as oxidant ?

(A) Na (B) Fe_2O_3 (C) Cl_2 (D) $CH_2 = CH_2$

5. Which of the elements show similarity when hydrogen gains one electron to attain stable electronic configuration like inert gas ?

(A) Alkali metal (B) Alkaline Earth metal (C) Halogen (D) Chalcogen

6. Which oxidation states are possible in hydrogen ?

(A) only 0 (B) +1, −1, 0 (C) only +1 (D) only −1

7. Which property of hydrogen differs from halogen ?

(A) Electronegative characteristics (B) Ionization enthalpy

(C) Non-metallic (D) Reductant

8. Which of the following statement is incorrect for protium, duterium and tritium ?

(A) They are isotopes of each other (B) They have 1, 2 and 3 nucleons respectively

(C) They have same electronic configuration, so physical properties are also same

(D) Their relative proportion of isotopes : 99.98 : 0.0516 : 10^{-15}

9. ${}^{3}_{1}H \rightarrow X + \beta^{-}$ particle, X = ...

(A) ${}^{2}_{1}H$ (B) ${}^{3}_{2}He$ (C) ${}^{1}_{1}H$ (D) ${}^{4}_{2}He$

Answers : 1. (C), **2.** (D), **3.** (A), **4.** (A), **5.** (C), **6.** (B), **7.** (D), **8.** (C), **9.** (B)

● **Preparation of dihydrogen**

Generally : $Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$

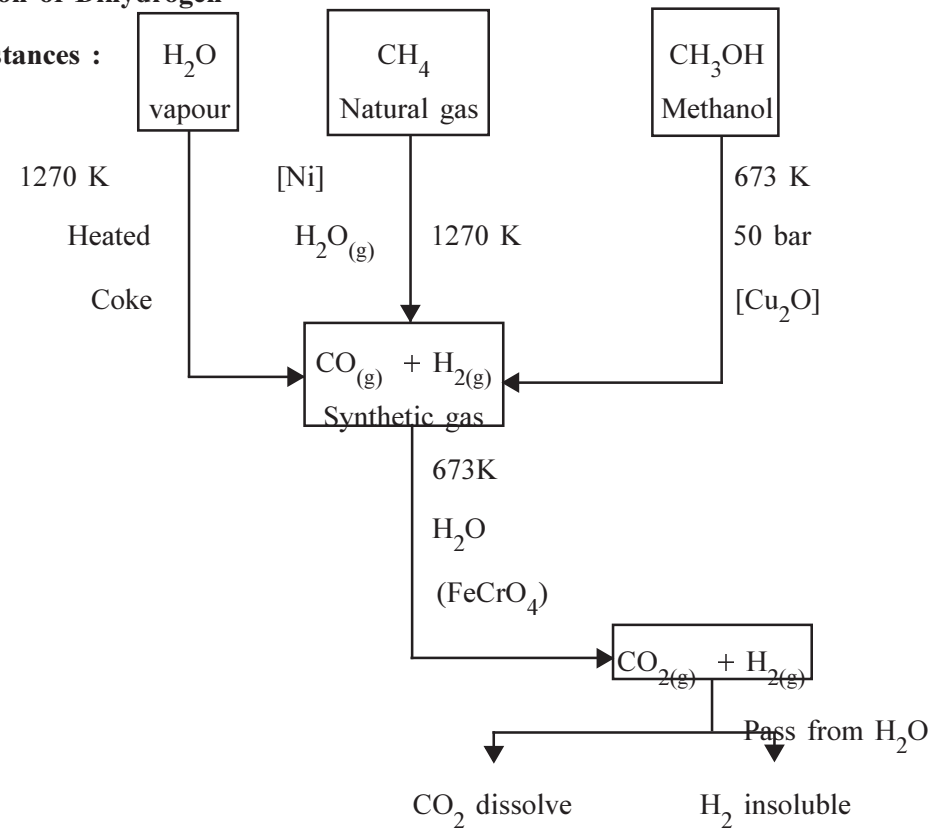
$Zn_{(s)} + 2NaOH_{(aq)} \rightarrow Na_2ZnO_{2(aq)} + H_{2(g)}$

Sodium zincate

Industrially : $Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}$

● **Industrial Preparation of Dihydrogen**

(i) Initial substances :



(ii) Electrolysis of water

	Electrolyte	Electrode
(a)	Acidic water	Pt
(b)	Aqueous solution of $Ba(OH)_2$	Ni

H_2 gas obtained on cathode.

● **Physical properties of Dihydrogen**

(1) Colourless, odourless, Tasteless gas
(2) Lighter than air
(3) Insoluble in water
(4) Diamagnetic
(5) Reduction potential 0.0 V

- **Chemical reactions of dihydrogen**

$H_{2(g)}$ + $X_{2(g)}$ → 2HX (X = F, Cl, Br, I)

$H_{2(g)}$ + $O_{2(g)}$ (catalyst, Δ) → $H_2O_{(l)}$ ΔH = −285.9 KJ/mole

$H_{2(g)}$ + $N_{2(g)}$ (773 K, 200 bar [Fe]) → $NH_{3(g)}$ ΔH = −92.6 KJ/mole

$H_{2(g)}$ + $2M_{(s)}$ → $2MH_{(s)}$ M = alkali metal

$H_{2(g)}$ + $Pd^{2+}_{(aq)}$ → $Pd_{(s)} + 2H^+_{(aq)}$

$H_{2(g)}$ + $M_xO_y(S)$ → $xM_{(s)} + yH_2O_{(l)}$

$H_{2(g)}$ + $CH_2=CH_2$ ([Ni] 390K) → $CH_3 - CH_3$ Ethane

$H_{2(g)}$ + $CH \equiv CH$ (Raney Ni, 390K) → $CH_3 - CH_3$ Ethane

$H_{2(g)}$ + $CO_{(g)}$ (573K, 50 bar, [Cu_2O]) → $CH_3OH_{(l)}$ Methanol

- **Hydride**

Noble gases, Indium (In), Thalium (Tl) and Group 7, 8 and 9 do not give hydride.
General formula : MH or M_mH_n.

Types of Hydride	Element	Properties	Uses
(1) Saline or Ionic	s-block Group-1 MH Group 2 MH_2	(i) Crystalline, solid, non volatile and non conductor (ii) $2H^- \xrightarrow{\text{anode}} H_{2(g)} + 2e^-$ (iii) Polymeric structure like $(BeH_2)_n$ and $(MgH_2)_n$ (iv) $MH_{(s)} + H_2O_{(l)} \rightarrow MOH_{(aq)} + H_{2(g)}$	Uses of LiH is to prepare $LiAlH_4$ and $LiBH_4$ which act as reducing agent
(2) Metallic or Intertitial	d-block, f-block, only Cr from 6th Group.	Non-stoichiometric because hydrogen is absorbed in the interstitial place of metal.	catalyst
(3) Molecular or Co-valent	Metallic and non metallic p-block elements, Electron deficient hydride : B_2H_6 Electron precise hydride : CH_4 Electron rich hydride : H_2O, NH_3, HF	(i) Gas, liquid form (ii) stable	

10. When Zn metal is added to concentrated NaOH it produces combustible gas and gives ..... .

(A) Na_2ZnO_3 (B) Na_3ZnO_2 (C) Na_2ZnO_2 (D) $NaZnO_2$

11. Which electrolysis gives pure dihydrogen ?

(A) Aqueous solution of KOH (B) Aqueous solution of $Ba(OH)_2$

(C) Water containing H_2SO_4 (D) Water containing NaOH

12. Water gas + $H_2O_{(g)} \xrightarrow[[Z]]{673K} CO_{2(g)} + 2H_{2(g)}$ Z = ...

(A) Cu_2O (B) Fe (C) Ni (D) $FeCrO_4$

13. Which of the following passes through $CO_{2(g)} + H_{2(g)}$ to get H_2 gas ?

(A) H_2O (B) Con. H_2SO_4 (C) dil. HCl (D) Hot solution of KCl

14. Which electrolysis produces H_2 gas on anode ?

(A) Aqueous solution of $Ba(OH)_2$ (B) Acidic water

(C) Fused sodium hydride (D) concentrated NaCl

15. Which of the following pair of elements do not give hydride compounds ?

(A) Ca, Be (B) In, Tl (C) Li, P (D) Sb, V

16. Which hydride of s-block possess covalent and polymeric structure ?

(A) BeH_2 (B) LiH (C) CaH_2 (D) NaH

17. $2MH + B_2H_6 \rightarrow 2MBH_4$ which block of M-element belong ?

(A) d (B) s (C) p (D) f

18. Position of element in periodic table, which of the following is electron precise hydride ?

(A) period 2, group 14 (B) period 2, group 15 (C) period 2, group 13 (D) period 2, group 16

19. What is the set of quantum no of valence electron in which it is electron deficient hydride ?

(A) n = 3, l = 1 (B) n = 2, l = 0 (C) n = 2, l = 1 (D) n = 4, l = 2

Answers : 10. (C), **11.** (B), **12.** (D), **13.** (A), **14.** (C), **15.** (B), **16.** (A), **17.** (B) **18.** (A), **19.** (C)

● **Water : Universal Solvent**

Physical Property :

(1) colourless, odourless, tasteless liquid

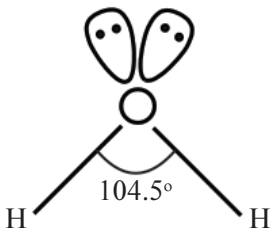
(2) Melting Point : 273K, boiling point : 373K, density 1.0 gm cm^{-3} at 298 K

(3) Polar, anomalous volume expansion between 273K to 277K

Structure :

(1) Types of hybridisation : sp^3

Bond angle : 104.5°, Bond length : 95.7 pm



Due to two non-bonding e^- pairs bond angle decreases from 109° 28'

(2) Ice : A definite three dimensional structure is formed by hydrogen bonds. It is lighter than water.

● Chemical Properties of Water

$H_2O_{(l)}$

- $\xrightarrow{\text{concentration}}$ 55.55 mol lit^{-1} at 298 K
- $\xrightarrow{H_2O}$ $H_3O^+_{(aq)} + OH^-_{(aq)}$ At equilibrium $[H_3O^+] = [OH^-] = 10^{-7}$M
- $\xrightarrow[\text{Base}]{NH_{3(g)}}$ $NH^+_{4\,(aq)} + OH^-_{(aq)}$
- $\xrightarrow[\text{Acid}]{H_2S}$ $HS^-_{(aq)} + H_3O^+_{(aq)}$ } Amphoteric nature
- $\xrightarrow[\text{Acid}]{HS^-}$ $S^{-2}_{(aq)} + H_3O^+_{(aq)}$
- $\xrightarrow{Na}$ $NaOH_{(aq)} + H_{2(g)}$ Oxidizing agent
- $\xrightarrow[\text{Photo synthesis}]{CO_2}$ $C_6H_{12}O_6 + O_2$ } Reducing agent
- $\xrightarrow{F_2}$ $HF_{(aq)} + O_{2(g)}$
- $\xrightarrow{P_4O_{10}}$ H_3PO_4
- $\xrightarrow{AlCl_3}$ $[Al(H_2O)_6]^{3+}_{(aq)} + 3Cl^-_{(aq)}$ } Hydration reaction
- $\xrightarrow{SiCl_4}$ $SiO_2 + HCl$
- $\xrightarrow{Ca_3N_2}$ $Ca(OH)_2 + NH_3$

Some compounds are hydrous. The number of molecules of water associated with it is called water of crystallization. Metallic salt : $CuSO_4 \cdot 5H_2O$, $FeSO_4 \cdot 7H_2O$, $BaCl_2 \cdot 2H_2O$

Complex salt : $[Ni(H_2O)_6]\,(NO_3^-)_2$, $[Cr(H_2O)_6]Cl_3$

The water which does not easily form leather with soap is called hard water.

$2C_{17}H_{35}COONa + M^{2+}_{(aq)} \rightarrow (C_{17}H_{35}COO)_2M_{(s)} + 2Na^+_{(aq)}$

(salt of fatty acid)

Hardness of water

- **Temporary**
  - Due to soluble $Mg(HCO_3)_2$ and $Ca(HCO_3)_2$
- **Permanent**
  - Due to soluble salt of $CaCl_2$, $CaSO_4$, $MgCl_2$ and $MgSO_4$

Methods of removal of hardness

(1) Boiling of water

$M(HCO_3)_{2(aq)} \xrightarrow{\Delta} MCO_{3(s)} + CO_{2(g)} + H_2O_{(l)}$

(2) Clark's method

$Ca(HCO_3)_{2(aq)} + Ca(OH)_{2(aq)}$ lime water $\downarrow$

$2CaCO_{3(s)} + 2H_2O_{(l)}$

● **Methods to remove permanent hardness**

(i) Chemical Method	(ii) Ion-Exchange	(iii) Synthetic Resin
(a) Add determined amount of washing soda $MCl_{2(aq)} + Na_2CO_{3(s)}$ $MCO_{3(s)} + 2NaCl_{(aq)}$ $M = Ca^{2+} / Mg^{2+}$	• uses of zeolite • It is aluminosilicate. $Na_2Al_2Si_4O_{12}$ • Structure like honey comb.	Very effective Exchange Ca^{2+} and Mg^{2+} from cation exchange region (RSO_3Na)
(b) By use of sodium hexa meta phosphate Ca^{2+} and Mg^{2+} ions can be made ineffective.	• Na^+ present in void exchange by Ca^{2+} and Mg^{2+} ion.	$2R\,Na_{(s)} + M^{+2}_{(aq)} \downarrow R_2M_{(s)} + 2Na^+_{(aq)}$
$Na_6P_6O_{18} \rightarrow 2Na^+ + Na_4P_6O_{18}^{-2}$ calgon $M^{2+} + Na_4P_6O_{18}^{-2} \rightarrow 2Na^+ +$ $[Na_2MP_6O_{18}]^{2-}$ (M = Ca, Mg).	$2Na^+Z^- + Ca^{2+} \rightarrow Ca^{2+}(Z^-)_2 + 2Na^+$	To remove anions from water, anion exchange resins are used. To make efficient resin again NaCl is passed through it.

20. H–O–H bond angle in water is 104.5° instead of 109° 28 ' because .....
(A) it possess sp^3 hybridisation. (B) it contains hydrogen bond
(C) it has non-bonding e-pair effect (D) it is polar

21. What is the concentration of pure water at 298 K temperature ?
(A) 10^{-7} M (B) 55.55 M (C) 10^{-14} M (D) 7.00 M

22. In the reaction : $H_2S_{(l)} + H_2O_{(l)} \rightleftharpoons HS^-_{(aq)} + H_3O^+_{(aq)}$ H_2O acts as .....
(I) Acid (II) Base (III) Reductant
(A) only (I) (B) (II) and (III) (C) (I) and (III) (D) only (II)

23. Hard water is not suitable for washing clothes because .....
(A) water forms micelle with soap (B) sodium salt of fatty acid is precipitated
(C) soap is hydrolysis by water (D) magnesium salt of fatty acid is precipitated

24. Which method is used to remove temporary hardness of water ?
(A) Distillation (B) Sublimation (C) Filteration (D) Boiling

25. Which substance is used for hard water becomes soft by Clark's method ?
(A) $CaCl_2$ (B) $Ca(OH)_2$ (C) $CaCO_3$ (D) $Ca(NO_3)_2$

26. Removal of permanent hardness of water calgon is used because .....
(A) to form soluble complex with $RCOO^-$ (B) precipitation of negative components
(C) to form soluble complex with Ca^{2+} and Mg^{2+} (D) precipitation of positive components.

27. Which method is used to soften hard water completely without salt ?
(A) Ion exchange (B) Electrolysis (C) Synthetic resin (D) Clark's method

Answers : 20. (C), **21.** (B), **22.** (D), **23.** (D), **24.** (D), **25.** (B), **26.** (C), **27.** (C)

- **Hydrogen Peroxide (H_2O_2)**

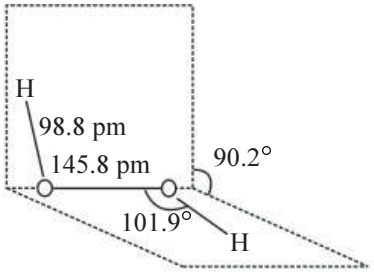
Physical Properties :

(i) H_2O_2 in pure form is colourless but is in more proportion, it is a blue coloured dense liquid.

(ii) Its density in solid form is 1.64 g cm^{-3} and in liquid form 1.44 gcm^{-3}.

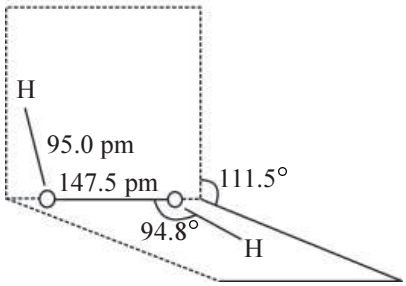
(iii) Melting Point = 272.4 K, boiling point = 423 K

- Strong oxidizing agent.
- H_2O_2 sold in market has H_2O_2 30% in 100 volume with water.
- At STP in one millilitre 30% H_2O_2 gives 100 ml of oxygen.



(Solid Form)

- **Structure**
  - There are two bonds O–O and H–O in H_2O_2.
  - In peroxide $(-O-O)^{2-}$ ion is present. Its molecule is non planar.



(Gas Form)

Uses :

(i) As bleaching agent.

(ii) As mild disinfectant in dentistry which is sold in market as perhydral.

(iii) In preparation of chemicals like sodium percarbonate, sodium perborate etc.

(iv) As the controller in pollutants.

- **Preparation and properties of H_2O_2**

(1) J. L. Thenard

$BaO_2 + H_2SO_4$
Barium peroxide

(2) Industrially

2–ethyl anthraquinol

O_2
Air

Oxidize

H_2/Pd
Oxidized

(3) High current density of HSO_4^-

Anodic oxidation

$H_2S_2O_8$ perdisulphuric acid

$-2HSO_4^-$

$-2H^+$

Hydrolysis

$2H_2O$

Hydrolysis

H_2O_2

Oxidant

$H_2O_2 \rightarrow H_2O$

$-1 \quad -2$

Reductant

$H_2O_2 \rightarrow O_2$

$-1 \quad 0$

$H_2SO_{4(aq)} + H_2O_{(l)} \xleftarrow{H_2SO_3}$

$HNO_{3(aq)} + H_2O_{(l)} \xleftarrow{HNO_2}$

$PbSO_{4(aq)} + H_2O_{(l)} \xleftarrow{PbS_{(s)}}$

$2Fe^{3+}_{(aq)} + 2H_2O_{(l)} \xleftarrow[2H^+]{2Fe^{2+}}$

$\xrightarrow{HOCl} H_3O^+ + Cl^- + O_{2(g)}$

$\xrightarrow{Cl_2+2OH^-} 2Cl^- + 2H_2O + O_{2(g)}$

$\xrightarrow{I_2+2OH^-} 2I^- + 2H_2O + O_{2(g)}$

$\xrightarrow{2Fe^{3+}+2OH^-} 2Fe^{2+} + 2H_2O + O_{2(g)}$

$\xrightarrow[KMnO_4]{Basic} MnO_2 + KOH + H_2O + O_2$

- **Heavy water (D_2O)**

  Oxides of dueterium

- **Chemical Properties**

  $SO_3 + D_2O \rightarrow D_2SO_4$,

  $CaC_2 + 2D_2O \rightarrow C_2D_2 + Ca(OD)_2$,

  $Al_4C_3 + 12D_2O \rightarrow 3CD_4 + 4Al(OD)_3$

- **Uses**

As moderator, to obtain compounds of dueterium to understand mechanism of a reaction exchange of ions etc.

- **Economic utility of Dihydrogen**

Because of decrease in quantity of crude oil available from the earth, not possible to store electrical energy and limitations of obtaining nuclear energy, the utility of dihydrogen can be the source of energy in its place

Hydrogen fuel is ecofriendly

- **Problems**

(i) production of low cost (ii) storage (iii) transformation

- **Uses of dihydrogen :**

(i) As a fuel in rocket (ii) As reductant (iii) Production of NH_3, CH_3OH, HCl etc. (iv) Used in cutting and welding work of metals.

- **Formulas of Examples**

(1) $M = \frac{W}{M' \times V}$ Where, M = molarity
W = weight of solute (gm.)
M' = molecular weight of solute (gm)
V = volume of solution (lit.)

(2) $N = \frac{W}{E \times V}$ Where, N = normality
W = weight of solute (gm.)
E = equivalent weight of solute
V = volume of solution (lit)

(3) $\frac{\text{gram}}{\text{litre}}$ = N × Equivulent weight

(4) Equivalent weight of H_2O_2 = 17

(5) N for H_2O_2 = 2M

(6) $\%\frac{W}{V}$ = gm of H_2O_2 dissolve in 100 ml volume of solution

(7) Volume = $\%\ \frac{W}{V} \times 3.294$

(8) $\%\frac{W}{V} = 3.4 \times M$

(9) % Volume = 11.2 × M

(10) $N_1V_1 = N_2V_2$

(11) $(\text{strength})_1 \times (\text{volume of solution})_1 + (\text{strength})_2 \times (\text{volume of solution})_2 + (\text{strength})_3 \times (\text{volume of solution})_3$ = Strength × Volume of mix solution

where, strength = $\%\frac{W}{V} = \frac{\text{gram}}{\text{litre}}$ = volume

28. Which statement is wrong according to H_2O_2 ?

(A) It is blue coloured dense liquid. (B) It contains two-OH group in one plane.

(C) It possess open book like structure. (D) As bleaching agent in leather industry.

29. Which behaviour of H_2O_2 show in the reaction ? $H_2O_{2(l)} \rightarrow 2H^+ + O_{2(g)} + 2e^-$

(A) oxidant (B) reductant (C) acidic (D) catalyst

30. Which of the following substance is reduced by H_2O_2 in basic medium ?

(A) Fe^{2+} (B) PbS (C) HOCl (D) SO_3^{2-}

31. Structure of H_2O_2 is ..... .

(A) planar (B) spherical (C) linear (D) non planar

32. Which product is obtained when calcium carbide reacts with heavy water ?

(A) CD_4 (B) CD_2 (C) Ca_2OD (D) C_2D_2

33. What litre of O_2 gas produced by decomposition of 40 gm litre of H_2O_2 at STP ?

(A) 9.52 litre (B) 22.4 litre (C) 13.17 litre (D) 17.13 litre

34. How many litres of O_2 gas is produced when 440 gm solute dissolved in 6 lit. H_2O_2 at STP ?

(A) 5.19 lit. (B) 24.156 lit. (C) 0.865 lit. (D) 144.9 lit.

35. What strength volume and % W/V of 48 gm solubility in 500 ml aqueous solution of H_2O_2 ?

(A) 31.62, 9.6 (B) 63.24, 19.2 (C) 3.162, 0.96 (D) 6.324, 1.92

36. What amount (gm) of 3.72 mol lit.$^{-1}$ aqueous solution of H_2O_2 in 8 lit volume ? As well as what strength in % W/V of its solution ?

	Amount (gm)	**% W/V**
(A)	505.92	12.648
(B)	1011.8	12.648
(C)	101.189	1.2648
(D)	50.592	1.2648

37. What is molarity (M), normality (N) and strength of solution in gm/lit respectively of 20 volume H_2O_2 solution ?

(A) 3.57, 3.57, 121.42 (B) 3.57, 7.14, 121.42 (C) 1.785, 3.57, 60.70 (D) 1.785, 1.78, 60.70

38. What is molarity (M), strength in gm/lit and volume of H_2O_2 solution, when 30 ml 0.1M $KMnO_4$ solution is required for complete oxidation of 30 ml acidic solution of H_2O_2 ?

(A) 0.12, 0.9, 0.6 (B) 0.06, 0.7, 0.8 (C) 0.1, 1.0, 0.9 (D) 0.05, 1.7, 0.56

39. What is molarity (M), normality (N), strength (%w/v), gm/lit and in volume of 10 lit H_2O_2 solution containing 284 gm H_2O_2 ?

	M	N	$\%\frac{W}{V}$	gm/ml	volume
(A)	1.67	1.67	5.68	56.78	18.70
(B)	1.67	3.34	2.839	5.678	18.70
(C)	0.835	1.67	2.839	28.39	9.352
(D)	0.835	0.835	1.419	28.39	9.352

40. What is strength in $\%\ \frac{W}{V}$, gm/lit and volume of solution of H_2O_2 prepared by mixing 400 ml 1.25% $\frac{W}{V}$, 350 ml 2.1% $\frac{W}{V}$ and 250 ml 2.65% $\frac{W}{V}$ strength of H_2O_2 ? As well as what is the molarity (M) and normality (N) of the resulting solution ?

	$\%\frac{W}{V}$	gm/lit	Volume	M	N
(A)	1.006	10.06	3.31	0.03	0.06
(B)	1.006	1.006	3.31	0.03	0.03
(C)	1.897	1.897	6.25	1.116	1.116
(D)	1.897	1.897	6.25	0.558	1.116

41. What is the normality of $KMnO_4$ solution, when 10 ml $KMnO_4$ aqueous solution is required for complete oxidation of acidic solution of 30 mol 1.5 volume H_2O_2 ?

(A) 0.4 (B) 0.19 (C) 0.8 (D) 0.65

Answers : **28.** (B), **29.** (B), **30.** (C), **31.** (D), **32.** (D), **33.** (C), **34.** (B), **35.** (A), **36.** (B), **37.** (C), **38.** (D), **39.** (C), **40.** (D), **41.** (C)

● **Column Type Questions**

42. Match column I with column II.

Column I (Substance)	Column II (Use)
(1) D_2O	(P) As a fuel
(2) H_2O_2	(Q) As moderator
(3) Liquid hydrogen	(R) As disinfectant
(4) $P_6O_{18}^{6-}$	(S) To understand mechanism of reaction
	(t) To make of Ca^{2+} and Mg^{2+} effective in hard water

(A) (1) → (Q), (2) → (R), (S), (3) → (P), (4) → (t)

(B) (1) → (Q), (R), (2) → (P), (3) → (S), (4) → (t)

(C) (1) → (Q), (S), (2) → (R), (3) → (P), (4) → (t)

(D) (1) → (Q), (P), (2) → (R), (S), (3) → (P), (4) → (t)

43. Match column I with column II.

Column I (Compounds)	**Column II (Types of hydride)**
(1) LaH_3	(P) Saline
(2) H_2S	(Q) Molecular
(3) CaH_2	(R) Electron rich
(4) B_2H_6	(S) Interstitial

(A) (1) → (Q), (2) → (R), (3) → (P), (4) → (Q), (R)
(B) (1) → (S), (2) → (Q), (3) → (P), (4) → (Q)
(C) (1) → (Q), (S), (2) → (R), (3) → (P), (4) → (R)
(D) (1) → (S), (2) → (Q), (R), (3) → (P), (4) → (Q)

44. Match column I with column II.

Column I	**Column II**
(1) Calgon	(P) Complex salt
(2) $BaCl_2 \cdot 2H_2O$	(Q) Market name of H_2O_2
(3) Perhydral	(R) $Na_4P_6O_{18}$
(4) Paladium hydride	(S) Metallic salt
	(T) $P_6O_{18}^{6-}$
	(U) Non-stoichiometric

(A) (1)→(R), (2)→(P), (3)→(Q), (4)→(S)
(B) (1)→(T), (2)→(S), (3→(Q), (4)→(U)
(C) (1)→(R), (2)→(P), (3)→(Q), (4)→(U)
(D) (1)→(T), (2)→(S), (3)→(Q), (4)→(P), (U)

Answers : 42. (C), **43.** (D), **44.** (B)

● **T = True, F = False Type Questions**

45. (i) To prevent the decomposition of aqueous solution of H_2O_2 stabilizer PO_4^{3-} is added.

(ii) H_2O_2 is useful to prepare good quality of detergent.

(iii) Use of H_2O_2 is dangerous to environment.

(iv) In basic medium H_2O_2 does not make colourless the aqueous solution of $KMnO_4$.

(A) TTTF (B) FTTT (C) TTFF (D) FTTF

46. (i) Dihydrogen is paramagnetic.

(ii) Ozone gas is used in cutting and welding of metals.

(iii) Hydrogenation of edible oil in presence of Ni catalyst to give edible fat.

(iv) Water gas is known as synthetic gas.

(A) FFTT (B) TFTT (C) TTTT (D) FFFT

47. (i) Na_2CO_3 is used to remove temporary and permanent hardness of water.

(ii) Basic solution containing Fe^{3+} ion becomes blue by addition of H_2O_2.

(iii) Oxidation number of hydrogen is +1 in saline hydride.

(iv) Percentage of deuterium in heavy water is 11.11.

(A) TTTF (B) FFFF (C) FFTF (D) FFTT

Answers : 45. (C), **46.** (A), **47.** (B)

- **There are two statements are given: Among this one is statement (A) and other is reason (R). Select correct option to study instructions carefully.**

  (A) Statement (A) and Reason (R) both are true and reason (R) is correct explanation of statement (A)

  (B) Statement (A) and Reason (R) both are true and reason (R) is not correct explanation of statement (A)

  (C) Statement (A) is correct but reason (R) is wrong.

  (D) Statement (A) and Reason (R) both are incorrect.

48. **Statement (A)** : Calgon is used to remove Ca^{2+} and Mg^{2+} from hard water.

Reason (R) : Calgon gives precipitate with Ca^{2+} and Mg^{2+}

49. **Statement (A)** : In acidic medium, H_2O_2, reacts with $KMnO_4$ to produce O_2.

Reason (R) : $KMnO_4$ oxidised of H_2O_2 in presence of O_2.

50. **Statement (A)** : Boron hydride is co-valent hydride.

Reason (R) : Difference of electronegativity is more between boron and hydrogen.

Answers : 48. (D), **49.** (A), **50.** (C).

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